

Easy Fence Brush

Official User Manual & Documentation

A non-destructive, spline-based placement tool for Unity Level Designers.

Version 1.0

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1 Introduction

Welcome to **Easy Fence Brush**, the ultimate spline-based placement tool designed for Unity Level Designers and Environment Artists. This tool eliminates the tedious process of manually placing repetitive objects. With an intuitive visual grid, smart terrain snapping, and procedural randomization, you can create complex, organic fences, walls, and barriers in seconds directly in the Unity Scene View.

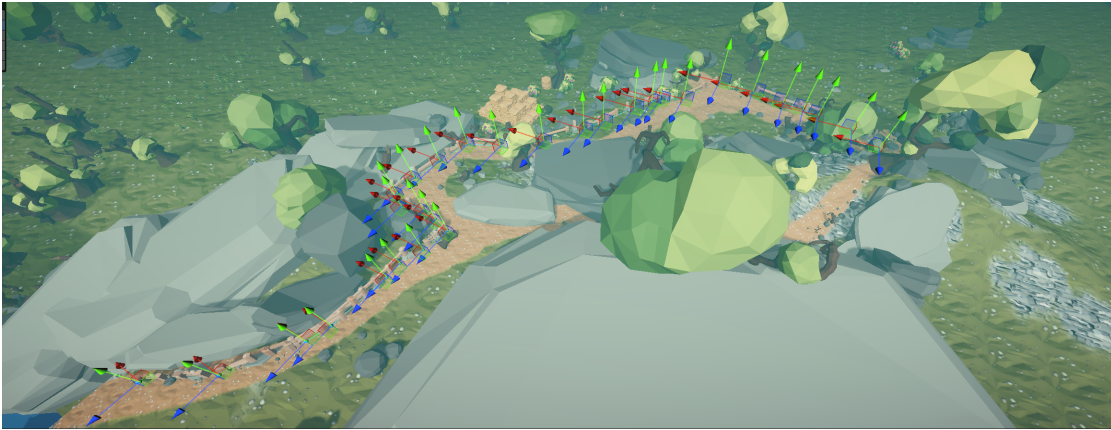


Figure 1: Easy Fence Brush overview.

2 Getting Started

2.1 Creating Your First Fence Brush

1. In the Unity Editor top menu, navigate to **Tools > Easy Fence > Create Brush**.
2. A new GameObject named `Fence Brush` will appear in your scene hierarchy, automatically equipped with the `EasyFenceTool` component.
3. Select the `Fence Brush` object to view the custom inspector.

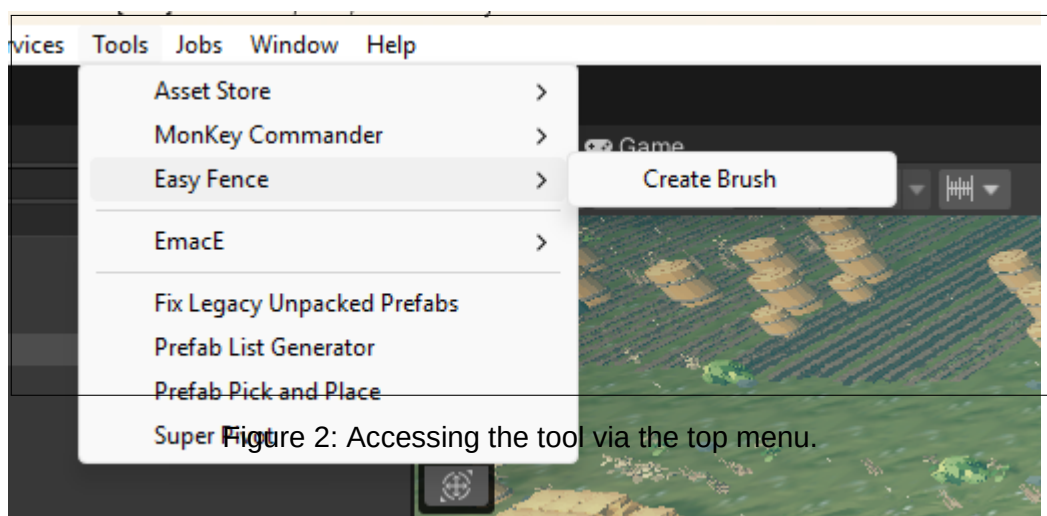


Figure 2: Accessing the tool via the top menu.

3 Drawing the Path

The core of Easy Fence Brush is its non-destructive path system. To interact with the path in the

Scene View, ensure the **EDITING MODE** is active (the button in the inspector will be highlighted

in green: **EDITING MODE ACTIVE**).

If you are starting from scratch, click the **[ADD INITIAL POINT]** button to spawn the first node exactly where your Scene View camera is looking.

3.1 Scene View Controls

- **Add a Point:** Hold Shift +Left Click on the terrain/surface to extend the fence.
- **Insert a Point:** Hold Shift +Left Click on the virtual line connecting two existing points. The tool will automatically split the segment and insert a new node.
- **Remove a Point:** Hold Ctrl +Left Click on an existing point (the node will turn red to indicate deletion).
- **Move a Point:** Click and drag the standard Unity position handle on any node to adjust its location.

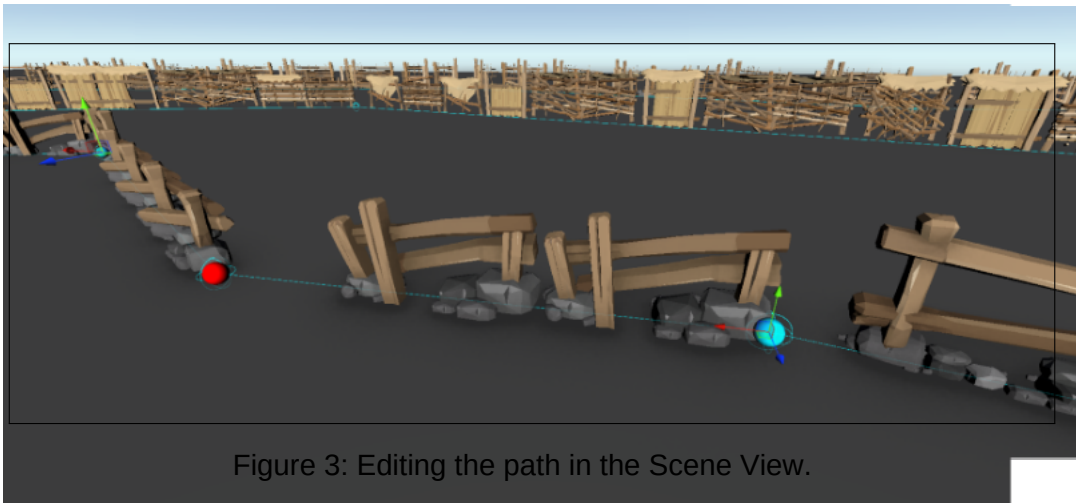


Figure 3: Editing the path in the Scene View.

3.2 Path Options

- **Is Loop:** Connects the last point of your path back to the first point, instantly creating a closed enclosure.
- **Insert Threshold:** Determines how close your mouse cursor needs to be to a line segment to insert a new point rather than adding one at the end of the path.

4 Managing Assets (Visual Grid System)

Easy Fence Brush separates your assets into two distinct categories: **Fence Segments** (the main body) and **Posts & Corners** (the pillars/nodes). Both sections utilize a highly visual, drag-and-drop workflow.

4.1 Importing Prefabs

You have two ways to feed assets into the brush using the toggle buttons:

1. **Manual List:** Drag and drop individual prefabs directly into the drop zone.
2. **Folder Directory:** Drag and drop a folder from your Project window. The tool will automatically fetch every prefab inside it.

4.2 Visual Asset Selection

Once imported, your prefabs will appear as a grid of thumbnail images.

- **Toggle On/Off:** Click any thumbnail to instantly include or exclude it from the generation pool.
- Disabled assets are grayed out. This allows you to quickly test different asset combinations without losing your references.
- Click the **Clear List** button to remove all assets, or **Enable All** to quickly restore excluded ones.

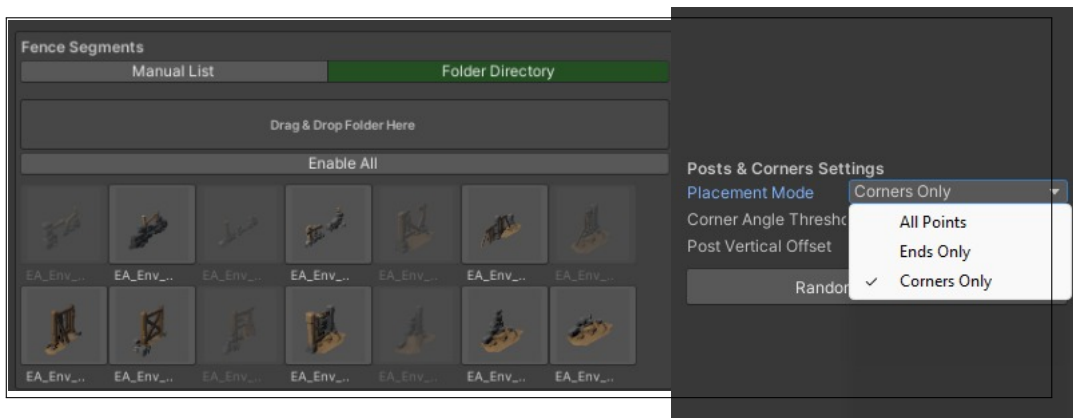


Figure 4: The visual asset grid.

5 Posts & Corners Settings

Posts are the connecting pillars between your fence segments. The tool offers intelligent logic to place them automatically where they make the most structural sense.

- **Placement Mode:**
 - *All Points*: Places a post at every single node you clicked.
 - *Ends Only*: Places posts only at the very first and very last point of the path.
 - *Corners Only (Smart Mode)*: Places posts at the ends, and additionally at path nodes where the angle bends sharply.
- **Corner Angle Threshold:** Used in *Corners Only* mode. Sets the minimum bend angle (from 1° to 180°) required to spawn a post.
- **Post Vertical Offset:** Allows you to independently raise or sink posts into the ground relative to the fence segments.

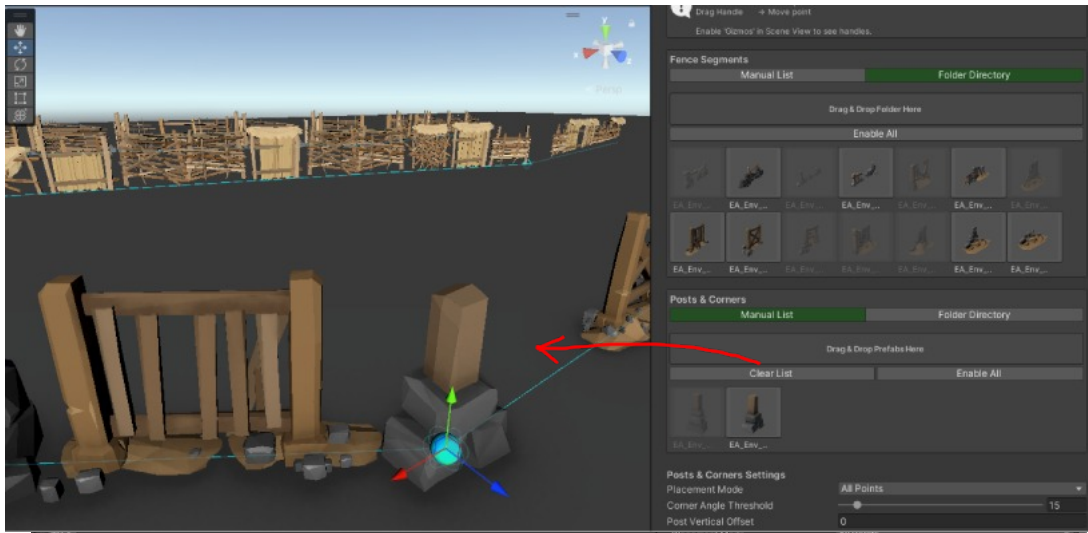


Figure 5: Post placement optimization.

6 Placement & Geometry

Achieve perfect alignment without mathematical guesswork.

- **Auto Calculate Length:** When enabled, the tool scans the mesh bounds of your active prefabs to determine their exact physical length dynamically. This guarantees that segments touch end-to-end perfectly.
- **Segment Length:** A manual override used ONLY if *Auto Calculate Length* is disabled.
- **Rotation & Position Offset:** Fine-tune the alignment if your models were exported with an unusual forward axis.

7 Terrain Adaptation

Make your fences look grounded and realistic, regardless of the topography.

- **Keep Vertical:** Forces the fence segments to stand straight up (parallel to global Y-axis), even when climbing steep hills.
- **Snap To Terrain:** Projects the path and all generated objects downwards to rest perfectly on the surface.
- **Snap Only To Unity Terrain:** Ignores standard 3D meshes (like houses) and only snaps to Unity Terrain components.
- **Vertical Offset:** A global height adjustment to sink the entire fence slightly into the dirt.

8 Natural Variation & Damage

Break up the repeating patterns of modular assets using procedural randomization to achieve a hand-crafted look.

- **Missing Segment Chance (0-1):** Introduces gaps. A value of 0.2 means every segment has a 20% chance of not spawning, creating a ruined look.

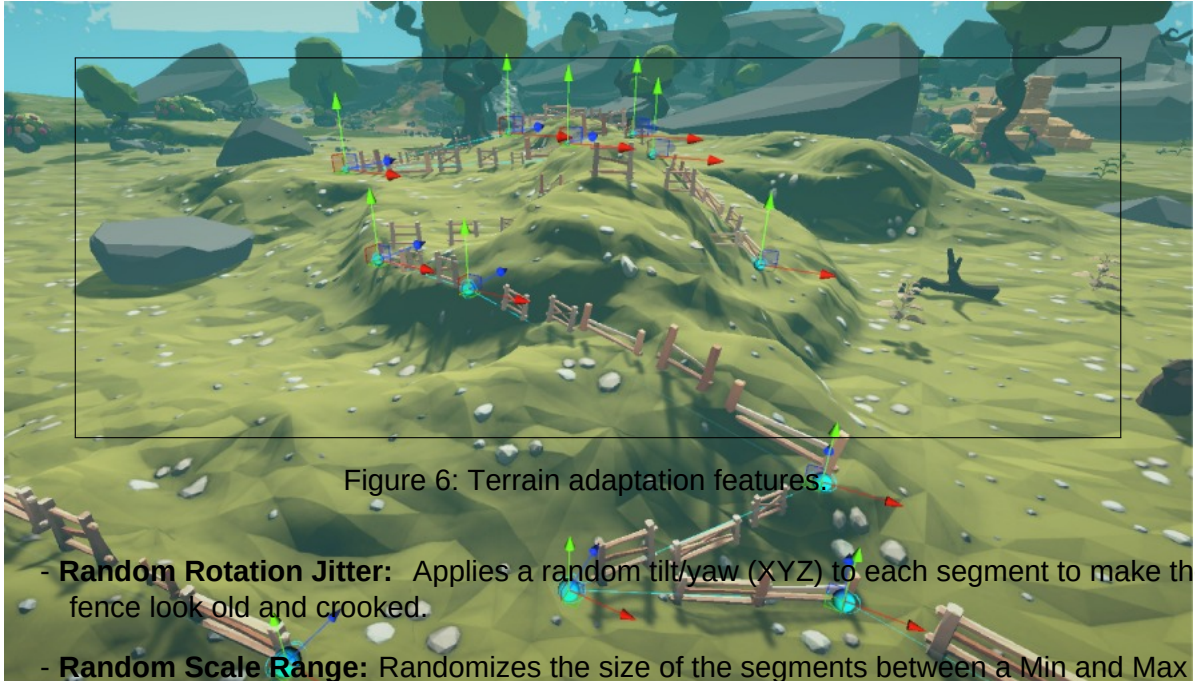
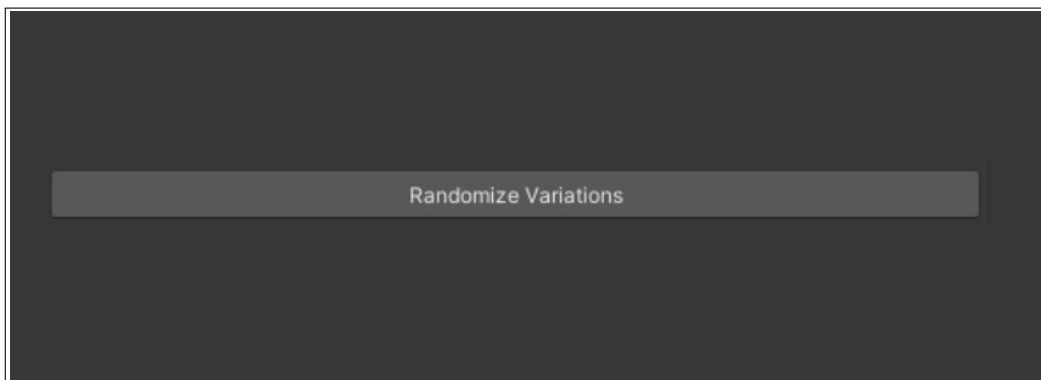


Figure 6: Terrain adaptation features.

- **Random Rotation Jitter:** Applies a random tilt/yaw (XYZ) to each segment to make the fence look old and crooked.
- **Random Scale Range:** Randomizes the size of the segments between a Min and Max multiplier.
- **Randomize Variations:** Click the **[Randomize Seed]** button to generate a new variation. The path stays the same, but gaps and jitters will re-roll.



9 Finalizing: Baking to Prefab

The fence you see while editing is a highly optimized, non-destructive **Preview**. It uses special Unity flags to keep your hierarchy clean and prevent scene bloat while you design.

Once you are completely satisfied with the shape, asset mix, and randomization:

1. Scroll to the very bottom of the inspector.
2. Click the green **[BAKE FENCE]** button.
3. A new, clean GameObject (e.g., Fence Brush_Baked) will be generated in your hierarchy, containing all the static mesh instances as child objects.
4. You can now safely delete the original Fence Brush, click **[Clear Path]** to start over, or move the brush to draw a new fence elsewhere.